- 1 Lab Objectives
  - 1 Set up a Jenkins controller
  - 2 Configure an agent
  - 3 Configure SSH Connection
  - 4 Run a Jenkins job on the agent
- 2 Requirements
  - 1 One machine for the Jenkins controller
  - 2 One or more machines for Jenkins agents
  - 3 SSH access between the controller and agent machines
- 3 Create VMs For Controller & Agent
  - 1 One VM From Controller
    - 1 Step 1: Create a new Security Group(Controller-SG) and add the following inbound rules
      - 1 SSH (Port 22) From Source 0.0.0.0/0
      - 2 Custom TCP (Port 8080) From Source 0.0.0.0/0
    - 2 Step 2: Launch an Ubuntu Linux Instance
      - 1 Image (AMI) Ubuntu Server 24.04 LTS
      - 2 Instance Type t2.micro
      - 3 Key Pair Select or create a key pair for SSH access to the instance
      - 4 Security Group Attach the Security Group you created in Step 1 Controller-Sg
  - 2 VM For Agent
    - 1 Step 1: Create a new Security Group (Agent-SG) and add the following inbound rules SSH (Port 22) From Source 0.0.0.0/0
    - 2 Step 2: Launch an Ubuntu Linux Instance
      - 1 Image (AMI) Ubuntu Server 24.04 LTS
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      - 4 Security Group Attach the Security Group you created in Step 1 - Agent-SG
- 4 Step-by-Step Instructions
  - 1 Step-1: Set Up the Controller
    - 1 Step 1: Install Java on Ubuntu
      - 1 Access Controller Node using SSH ssh -i <private-key.pem> ubuntu@<Controller Public IP>
      - 2 Login as Root sudo -i
      - 3 Update the package list sudo apt update
      - 5 Install Java 17 sudo apt install openjdk-17-jre-headless
      - 6 Verify Java installation java -version
    - 2 Step 2: Jenkins Installation
      - 1 Access Controller Node Again using SSH ssh -i <private-key.pem> ubuntu@<Controller Public IP>
      - 1 Step 1 Login as Root sudo -i
      - 2 Step 2 Download the Jenkins Repository Key wget -O /usr/share/keyrings/jenkins-keyring.asc \ https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
      - 3 Step 3 Add the Jenkins repository to your package manager echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc]" \ https://pkg.jenkins.io/debian-stable binary/ | sudo tee \ /etc/apt/sources.list.d/jenkins.list > /dev/null
      - 4 Step 4 Update Packages and Install Jenkins
        - 1 apt update
        - 2 apt install jenkins -y
      - 5 Step 5 Start the Jenkins service systemctl start jenkins
      - 6 Step 6 Enable Jenkins to start at boot systemctl enable jenkins
    - 3 Step 3: Access Jenkins
      - 1 Step 1 Get your EC2 instance's public IP from your cloud console
      - 2 Step 2 Open a web browser and go to http://Controller-Node-public-ip:8080
      - 3 Step 3 Unlock Jenkins
        - 1 Jenkins will ask for an initial admin password to unlock it
        - 2 Retrieve this password by running cat /var/lib/jenkins/secrets/initialAdminPassword
        - 3 Copy the password and paste it into the Jenkins setup wizard in your browser
        - 4 Choose Install Suggested Plugins Wait till Plugins Install Finish
        - 5 Create First Admin User Skip and Continue As Admin
        - 6 Save and Finish
        - 7 Start Using Jenkins
        - 8 Change Admin Password
          - 1 From Dashboard Manage Jenkins
          - 2 Users admin Settings Icon
          - 3 Security <Set New Password>
          - 4 Save
          - 5 Sign in to Jenkins
            - 1 Username Admin
            - 2 Password <Your Password>
  - 2 Step-2: Set Up the Agents
    - 1 Access Agent Node using ssh ssh -i <private-key.pem> ubuntu@<Agent Public IP>
    - 2 Step 2: Install Java on Ubuntu
      - 1 Update the package list sudo apt update
      - 2 Install Java 17 sudo apt install openjdk-17-jre-headless
      - 3 Verify Java installation java -version
  - 3 Step 3: Configure SSH
    - 1 Step:1 Controller SSH setup
      - 1 Access Controller Node using SSH
      - 2 Create User Name jenkins sudo useradd -m -d /home/jenkins -s /bin/bash jenkins
      - 3 Set password sudo passwd jenkins
      - 4 Login as jenkins su - jenkins
      - 5 Generate the key pair on controller
        - 1 ssh-keygen -t rsa -b 2048
          - 1 This command will generate 2 Key Public & Private
          - 2 Public key name id_rsa.pub
          - 3 Private key name id_rsa
        - 2 Press Enter to accept the default location (~/.ssh/id_rsa)
        - 3 Enter passphrase No need to enter password press Enter two time
      - 6 Copy the content of id_rsa.pub
        - 1 cat ~/.ssh/id_rsa.pub
        - 2 Output looks like this ssh-rsa AAAAB3...controller@hostname
        - 3 Copy the output of this command into notepad
    - 2 Step:2 Add the Public Key to the Agent Node
      - 1 Access Controller Node using SSH
      - 2 Create User Name jenkins sudo useradd -m -d /home/jenkins -s /bin/bash jenkins
      - 3 Set password sudo passwd jenkins
      - 4 Login as jenkins su - jenkins
      - 5 Append the public key (copied from the controller) to the agent's authorized_keys file
        - 1 Make sure you in /home/jenkins pwd
        - 2 create directory mkdir .ssh
        - 3 echo "<controller-public-key>" > ~/.ssh/authorized_keys
        - 4 Example: If your public key is ssh-rsa AAAAB3...controller@hostname
      - 6 Set Correct Permissions Ensure the .ssh directory and authorized_keys file have the correct permissions
        - 1 chmod 700 ~/.ssh
        - 2 chmod 600 ~/.ssh/authorized_keys
      - 7 Test the SSH Connection
        - 1 ssh jenkins@<agent-private-ip>
        - 2 If everything is set up correctly, you should log in without being prompted for a password
      - 8 Note The public key from the controller is now recognized by the agent, allowing the controller to SSH into the agent securely and without requiring a password
  - 3 Step 4: Add Agent Nodes to Jenkins Controller
    - 1 Step:1 Open the Jenkins Dashboard http://<controller-ip>:8080
    - 2 Step:2 Add a New Node
      - 1 Go to Manage Jenkins
      - 2 Nodes
      - 3 Click + New Node
    - 3 Step:3 Provide the following detail
      - 1 Node Name Agent-2
      - 2 Select Permanent Agent
      - 3 Click Create
  - 4 Step 5: Configure the Node
    - 1 Step:1 Set the Node Details
      - 1 Remote root directory
        - 1 Set the directory on the agent where Jenkins will store its files
        - 2 /home/jenkins
      - 2 Add descriptive labels for the node python-test
      - 3 Usage
        - 1 Use this node as much as possible For general-purpose agents
        - 2 Only build jobs with label expressions matching this node For specific jobs only
    - 2 Step:2 Launch Method Select Launch agent by SSH
    - 3 Step:3 Enter Connection Details
      - 1 Host Enter the agent node's IP address or hostname
      - 2 Credentials Add SSH credentials
        - 1 Click Add Jenkins
        - 2 select SSH Username with Private Key
        - 3 Copy Private Key From Controller
          - 1 Go to controller node
          - 2 Make sure you are login as jenkins user
          - 3 Open private key file cat .ssh/id_rsa
          - 4 Copy the content of private key file
            - 1 Always double-check that the private key contain starts with -----BEGIN and ends with -----END
            - 2 If the first or last line is missing, the key will become invalid, and you will encounter errors when attempting to use it for SSH connections or other operation
        - 4 Enter the username
          - 1 jenkins
          - 2 Paste the private key (id_rsa) from your controller
        - 5 Host Key Verification Strategy Manually Trusted key verification strategy
        - 6 Port Leave as 22 (default SSH port)
      - 3 Advanced Options (if needed) Adjust the SSH timeout or retry settings if you experience connection issues
      - 4 Click Save and Apply
  - 5 Step 6: Verify the Connection
    - 1 Go to Manage Jenkins > Node
    - 2 Check the agent's status
      - The agent is online and will accept tasks for execution when it has free executors.
      - The agent has been explicitly set to temporarily offline by a user. If the agent is still connected any running tasks will finish. No new tasks will start until the agent is put online again by a user.
      - The agent is not accepting tasks for execution. This usually means that the agents availability settings prevent task execution, e.g. when using `Bring this agent online according to a schedule` and being outside the time window. The agent might still execute tasks that have started while it was accepting tasks. Cloud agents that are intended to run only a single task might also use this. Agents will be shown as `suspended` in the `Build Executor Status` widgets.
      - None of the above conditions applies. The agent is not connected or it is still connected but a node monitor decided that the agent is not in a healthy state. An agent in this state
      - A green checkmark
- 5 Run a Job on the Agen
  - 1 Create a new job
    - 1 In the Jenkins dashboard, click New Item and select Freestyle Project
    - 2 Configure the job to execute a simple script (e.g., echo "Hello from the Agent!")
  - 2 Assign the job to the agent: Under Build Environment, specify the agent label
  - 3 Build the job Click Build Now and monitor the console output to confirm the job runs on the agent

Jenkins Controller-Agent Hands-On (SSH)